QP CODE: 25901162

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU – UGP (HONOURS) REGULAR EXAMINATION March 2026

FOURTH SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCCND202 - THERAPEUTICNUTRITION

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 50

Remember(K),Understand(U),Apply(A),Analyse(An),Evaluate(E),Create(C),Interest (I),Appreciation(Ap),and Skill(S)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple ChoiceQuestions

Answer all questions

Each question carries 1 mark

1. b). Increased energy needs
2. b). Dehydration
3. b). Immune-mediated adverse reaction to food
4. c). Anaphylaxis
5. a). Low fiber diet
6. b). Final stage of HIV infection

Part B

Short Answer

Answer any 2 questions.

Each question carries 2 marks.

7. What are the two causes of fever? [U] / [CO1]

Infectious causes – due to microorganisms (bacteria, viruses, fungi, parasites). *(1 mark)*

Non-infectious causes – due to inflammation, allergy, drugs, injury, or other diseases. *(1 mark)*

8. Name one disease associated with chronic fever. [U] / [CO1]

- Tuberculosis *(example of a disease causing chronic fever)* – 1 mark
- Any one correct disease mentioned (e.g., Malaria, Typhoid fever, etc.) – 1 mark

9.Name two foods that should be avoided in GERD. [U] / [CO2]

- Spicy foods – 1 mark
- Fried foods – 1 mark

(Any other correct examples like Caffeine or Chocolate can also be awarded marks.)

10. State the purpose of an elimination diet. [K] / [CO3]

- To identify foods that trigger symptoms/allergies or intolerances – 1 mark
- By systematically removing and then reintroducing specific foods to observe reactions – 1 mark

Part C

Short Essay

Answer any 4 questions

Each question carries 5 marks.

11. Write a short note on allergenic beverages. [U] / [CO3]

- Definition: Beverages that can trigger allergic reactions or intolerance in sensitive individuals – 1 mark
- Common examples: Milk, Soy milk, beverages containing Caffeine (coffee, tea), chocolate drinks – 1 mark

- Allergenic components: Proteins (milk, soy), additives, preservatives, or stimulants like caffeine – 1 mark
- Symptoms caused: Skin rashes, gastrointestinal upset, breathing difficulty, headache – 1 mark
- Management: Avoidance of trigger beverages, reading labels, use of hypoallergenic alternatives – 1 mark

12. Explain the aetiology and symptoms of upper gastrointestinal tract disorders. [U] / [CO2]

Aetiology (Causes) – *2–3 marks*

- Infection (e.g., *H. pylori* in gastritis/ulcer) – 1 mark
- Dietary factors – spicy foods, alcohol, irregular meals – 1 mark
- Stress and lifestyle factors (smoking, poor eating habits) – ½–1 mark
- Drug use (NSAIDs, steroids) – ½–1 mark
- Excess acid secretion / reflux (e.g., Gastroesophageal Reflux Disease) – ½–1 mark

Symptoms – *2–3 marks*

- Epigastric pain or burning sensation – 1 mark
- Nausea and vomiting – 1 mark
- Heartburn and acid regurgitation – 1 mark
- Loss of appetite, bloating, indigestion – ½–1 mark
- In severe cases: bleeding, weight loss – ½–1 mark

13. Describe plant-source food allergens. [U] / [CO3]

- Definition: Plant-source food allergens are allergenic substances present in foods derived from plants that can trigger immune reactions in sensitive individuals – 1 mark
- Common sources:
  Peanuts, Soybeans, Wheat, tree nuts (e.g., Almonds) – 1 mark
- Allergenic components:
  Proteins such as storage proteins, gluten (in wheat), and other plant proteins – 1 mark
- Symptoms caused:
  Skin reactions (rash, itching), gastrointestinal issues, respiratory problems, and in severe cases anaphylaxis – 1 mark
- Management:
  Avoidance of allergenic foods, proper food labeling awareness, substitution with hypoallergenic alternatives – 1 mark

14. Explain macronutrient deficiencies in AIDS. [U] / [CO1]

- Definition/Overview:
  In Acquired Immunodeficiency Syndrome, macronutrient deficiencies occur due to increased metabolic demands, poor intake, and malabsorption – 1 mark
- Causes:
  Reduced appetite, infections, malabsorption, drug side effects, and increased energy/protein needs – 1 mark
- Protein deficiency:
  Leads to muscle wasting, weight loss, weakened immunity (cachexia) – 1 mark
- Energy (carbohydrate & fat) deficiency:
  Causes fatigue, severe weight loss, and loss of body fat stores – 1 mark
- Consequences/Management:
  Leads to wasting syndrome; managed by high-calorie, high-protein diet, nutritional support, and treatment of infections – 1 mark

15. Explain the dietary guidelines for managing gastritis. [U] / [CO2]

Dietary guidelines for managing gastritis

- Eat small, frequent meals – avoids excess gastric acid secretion and reduces stomach irritation *(1 mark)*
- Avoid irritant foods – such as spicy, fried, oily, and highly seasoned foods, which increase gastric inflammation *(1 mark)*
- Limit gastric acid stimulants – avoid caffeine, alcohol, and carbonated beverages *(1 mark)*
- Include bland, easily digestible foods – such as boiled rice, soft cereals, curd, and well-cooked vegetables *(1 mark)*
- Maintain healthy eating habits & lifestyle – regular meal timing, avoid skipping meals, reduce stress, and avoid smoking *(1 mark)*

16. Describe the complications of AIDS. [U] / [CO1]

Complications of Acquired Immunodeficiency Syndrome

- Opportunistic infections – due to weakened immunity (e.g., tuberculosis, pneumonia, fungal infections) – 1 mark
- Cancers – increased risk of malignancies such as Kaposi's sarcoma and lymphoma – 1 mark
- Severe weight loss (wasting syndrome) – progressive loss of body weight and muscle mass – 1 mark
- Neurological complications – memory loss, confusion, neuropathy, and dementia – 1 mark
- Organ failure / multi-system involvement – chronic damage to lungs, liver, and other organs leading to death if untreated – 1 mark

Part D

Essay

Answer any 2 questions

Each question carries 10 marks

17. Explain lower gastrointestinal tract disorders with causes and symptoms. [U] / [CO2]

A. Definition (1 mark)

- Disorders affecting the lower gastrointestinal tract (small intestine, large intestine, rectum, anus) that interfere with digestion, absorption, and elimination – *(1 mark)*

B. Causes (4 marks)

- Infections (bacterial, viral, parasitic infections from contaminated food/water) – *(1 mark)*
- Dietary factors (low fiber diet, excessive junk food, irregular eating habits) – *(1 mark)*
- Inflammatory and functional disorders such as Inflammatory bowel disease and irritable bowel conditions – *(1 mark)*
- Other factors: stress, medications, and altered gut flora – *(1 mark)*

C. Symptoms (4 marks)

- Altered bowel habits: diarrhea or constipation – *(1 mark)*
- Abdominal pain and cramping – *(1 mark)*
- Bloating and flatulence – *(1 mark)*
- Severe signs: blood in stool, weight loss, dehydration, weakness – *(1 mark)*

D. Conclusion (1 mark)

- Early diagnosis, dietary modification, and treatment of underlying cause are essential to prevent complications – *(1 mark)*

18. Explain biochemical and immune tests used in food allergy diagnosis. [U] / [CO3]

Biochemical and immune tests used in food allergy diagnosis

A. Biochemical tests (5 marks)

- Serum histamine level test – Measures histamine released during allergic reaction *(1 mark)*
- Tryptase level test – Elevated in mast cell activation/anaphylaxis *(1 mark)*
- Eosinophil count (CBC) – Increased eosinophils indicate allergic response *(1 mark)*
- IgE-related biochemical estimation (total IgE) – Indicates allergic tendency *(1 mark)*

- Mediator release tests (e.g., leukotrienes) – Detect inflammatory mediators in allergy *(1 mark)*

B. Immune (immunological) tests (5 marks)

- Skin prick test (SPT) – Allergen applied to skin; wheal indicates IgE-mediated allergy *(1 mark)*
- Specific IgE test (RAST / ImmunoCAP) – Measures allergen-specific IgE antibodies in blood *(1 mark)*
- Elimination diet and oral food challenge – Gold standard for identifying trigger foods *(1 mark)*
- Basophil activation test (BAT) – Measures activation of basophils on exposure to allergens *(1 mark)*
- Patch test (for delayed allergy reactions) – Detects non-IgE mediated hypersensitivity *(1 mark)*

19. Explain malaria with causes, symptoms and types of fever. [U] / [CO1]

Malaria – causes, symptoms and types of fever

A. Definition (1 mark)

- Malaria is a mosquito-borne infectious disease caused by *Plasmodium* parasites and transmitted by female *Anopheles* mosquito.

B. Causes / Etiology (3 marks)

- Infection by *Plasmodium* species (e.g., *P. vivax, P. falciparum, P. malariae, P. ovale*) – 1 mark
- Transmission through bite of infected female *Anopheles* mosquito – 1 mark
- Parasite multiplies in liver and red blood cells causing fever episodes – 1 mark

C. Symptoms (3 marks)

- High fever with chills and rigor – 1 mark
- Headache, body ache, fatigue, and sweating – 1 mark
- Nausea, vomiting, anemia, and splenomegaly – 1 mark

D. Types of fever pattern in malaria (3 marks)

- Tertian fever – fever every 48 hours (seen in *P. vivax, P. ovale*) – 1 mark
- Quartan fever – fever every 72 hours (seen in *P. malariae*) – 1 mark
- Irregular or malignant fever – seen in *P. falciparum*, often severe and continuous – 1 mark.

20. Discuss the dietary management of diarrhoea. [U] / [CO2]

A. Definition / Aim of diet (1 mark)

- Dietary management of diarrhoea aims to replace fluids and electrolytes, reduce intestinal irritation, and restore normal bowel function – *(1 mark)*

B. Fluid and electrolyte replacement (2 marks)

- Increase intake of oral rehydration solutions (ORS), coconut water, clear soups – *(1 mark)*
- Maintain hydration to prevent dehydration and electrolyte imbalance (sodium, potassium loss) – *(1 mark)*

C. Energy and protein intake (2 marks)

- Provide easily digestible, high-energy foods like rice, banana, potatoes – *(1 mark)*
- Include moderate protein sources like curd, dal water, eggs (if tolerated) – *(1 mark)*

D. Foods to include (2 marks)

- Low-fiber, bland diet (BRAT diet: banana, rice, apple, toast) – *(1 mark)*
- Cooked cereals, boiled vegetables, yogurt/probiotics to restore gut flora – *(1 mark)*

E. Foods to avoid (2 marks)

- Avoid spicy, fatty, fried foods and caffeine (irritate gut) – *(1 mark)*
- Avoid milk (in lactose intolerance), raw vegetables, carbonated drinks – *(1 mark)*

F. Feeding guidelines / general advice (1 mark)

- Give small frequent meals, maintain hygiene, and gradually return to normal diet after recovery – *(1 mark)*